Worksheet 1

1. $y' = e^{2x} \cdot 2$
2. $f'(x) = \cos(e^{3x}) \cdot e^{3x} \cdot 3$
3. $y' = e^{\sin(x^4)} \cdot \cos(x^4) \cdot 4x^3$
4. $f(t) = t^{-4}$, $f'(t) = -4t^{-5} = \frac{-4}{t^5}$
5. $y' = 10(t-t^3)^9 \cdot (1-3t^2)$
6. $f'(t) = 4\cos^3 3t \cdot (-\sin 3t) \cdot 3$
7. $f'(x) = 2^x \ln 2$
8. $f'(x) = 2^{\operatorname{cosec}(8x)} \cdot \ln 2 \cdot (-\operatorname{cosec}(8x)\cot(8x)) \cdot 8$
9. $f'(x) = -\operatorname{cosec}^2(2^{1-5x}) \cdot 2^{(1-5x)} \cdot \ln 2 \cdot (-5)$
10. $f'(t) = 7\sec^2(3t) \cdot 3 - 3^{\tan 7t} \cdot \ln 3 \cdot \sec^2 7t \cdot 7$
11. $f'(x) = 0$
12. $f'(x) = \dfrac{\frac{1}{2}(x^3+1)^{-1/2} \cdot 3x^2(7-\tan 7x) - \sqrt{x^3+1}\,(-\sec^2 7x \cdot 7)}{(7-\tan 7x)^2}$

Worksheet 2

1. $y' = 4\left(\frac{1+x}{x^3}\right)^3 \cdot \dfrac{1(x^3)-(1+x)\cdot 3x^2}{(x^3)^2}$
2. $y' = \dfrac{0 - 3(2-3x)^2 \cdot -3}{[(2-3x)^3]^2} = \dfrac{+9}{(2-3x)^4}$
3. $f'(t) = -\operatorname{cosec}^2(2^t) \cdot 2^t \cdot \ln 2$
4. $f'(t) = \sec^2(e^t) \cdot e^t(e^{\tan t}) + \tan(e^t) \cdot e^{\tan t} \cdot \sec^2 t$
5. $f'(v) = \frac{1}{2}(v^3+\sec v)^{-1/2}(3v^2+\sec v \tan v)$
6. $f'(s) = \dfrac{3}{(1-3s)^2}$

7. $f'(x) = -2\sin(e^{-4x}) \cdot e^{-4x} \cdot -4$

8. $f'(x) = (e^{\sin 4x} \cdot \cos 4x \cdot 4)\cos(2x) + e^{\sin 4x} \cdot (-\sin(2x) \cdot 2)$

9. $f'(x) = 2^{\tan 3x} \cdot \ln 2 \cdot \sec^2 3x \cdot 3$

10. $f'(x) = 5\sec(3^x)\tan(3^x) \cdot 3^x \cdot \ln 3$

11. $f'(x) = 0$

12. $f'(x) = 3(1+2\sin 5x)^2(2\cos 5x \cdot 5)$

13. $f'(x) = 3^{5x} \cdot \ln 3 \cdot 5 + 15x^2 + 5^{3x} \cdot \ln 5 \cdot 3$

14. $f'(x) = 50(1-2x)^9 \cdot (-2)$

15. $f'(x) = \dfrac{4\cos(x^4) 4x^3 (e^{4\sin x}) - 4\sin(x^4) e^{4\sin x} \cdot 4\cos x}{(e^{4\sin x})^2}$

16. $f'(x) = \frac{1}{2}(\cos(e^{3x+1}))^{-1/2} \cdot -\sin(e^{3x+1}) \cdot e^{3x+1} \cdot 3$

17. $y' = 2e^{\cot x} \cdot -\operatorname{cosec}^2 x - 3\operatorname{cosec}^2(e^{-x}) \cdot e^{-x} \cdot -1.$

18. $f'(x) = \frac{1}{3}(\sec(4x))^{-2/3} \cdot \sec(4x)\tan(4x) \cdot 4\,(\tan(x^4)) +$
$\sqrt[3]{\sec(4x)} \cdot \sec^2(x^4) \cdot 4x^3$

19. $f'(x) = \dfrac{\frac{1}{2}(x^3+1)^{-1/2} \cdot 3x^2(7-\tan 3x) - \sqrt{x^3+1}\,(-\sec^2 3x \cdot 3)}{(7-\tan 3x)^2}$

20. $$y' = \frac{\left[(2^{3x} \cdot \ln 2 \cdot 3)(\sin 4x) + 2^{3x}\sin(4x) \cdot 4\right]\cos(1-x) \cdot \operatorname{cosec}(e^{-x}) - 2^{3x}\sin(4x)\left[+\sin(1-x)\operatorname{cosec}(e^{-x}) + \cos(1-x) \cdot -\operatorname{cosec}(e^{-x})\cot(e^{-x}) \cdot e^{-x} \cdot -1\right]}{\left[\cos(1-x)\operatorname{cosec}(e^{-x})\right]^2}$$

What a mess, maybe we should try something else with (20).?